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Periodic surface structures induced by femtosecond laser single pulse and pulse trains on metals

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Abstract
We experimentally investigate the formation of nanostructures on the surfaces of aluminum and copper with femtosecond laser single pulse and pulse trains. Subwavelength ripples were obtained on the surface of aluminum. The ripples' period gradually increases as the laser pulse fluence increases. With a single pulse ablation of copper, interlaced ripple structures formed on the wall of the crater when the laser pulse number is 2000, the reason for interlaced ripple formation is also discussed. A large scale of ripple structures formed by using a femtosecond laser single pulse and pulse trains to scan copper. We suggest that a pulse train is more effective in obtaining high quality ripple structures compared with a femtosecond laser single pulse.

Keywords: femtosecond laser, pulse train, surface structures, metal

(Some figures may appear in colour only in the online journal)

1. Introduction

Over the last few years, ultrashort laser pulses have proven to be a very promising tool for processing solid materials, especially metals, dielectrics and semiconductors [1–6]. Compared with long-pulse lasers, a femtosecond (fs) laser can enhance the machining efficiency of solid materials due to its ultrashort pulse durations and ultrahigh power densities. Because of the very short heating duration and rapid solid-to-vapor phase transition, the amount of thermal diffusion can be limited. This makes it possible to limit the absorption region to the penetration of the optical pulse thus limiting the collateral damage and considerably increasing the fabrication precision and quality [7–9]. Furthermore, a shaped pulse can control electron dynamics due to the duration of shaped pulse being much shorter than most of the physical characteristics [1, 10]. So far, a large amount of experimental studies have been performed to investigate the effect of temporally shaped pulses on the morphology of metallic surfaces [11–14]. This offers new possibilities for manipulating the transient localized material properties and corresponding phase-change mechanisms, which makes it much easier to optimize the laser parameters to achieve the desired ablation results for many applications [1, 15–17].

In this work, we performed an evolution of nanostructure morphologies induced by a fs laser on the surfaces of aluminum (Al) and copper (Cu). The behavior of the induced nanostructures presented differently with changes to the laser parameters, such as laser fluence, pulse number and pulse separation. Mechanisms related to nanoripple formation and its potential applications are also discussed.

2. Experimental setup

The schematic diagram of the experimental setup used for this work is depicted in figure 1. A Spectra Physics Spitfire regenerative amplifier producing 800 nm, 35 fs pulses at a repetition rate of 1 KHz with a pulse energy up to ~3 mJ was used. The fs laser pulse was linearly polarized. The energy of the laser pulse

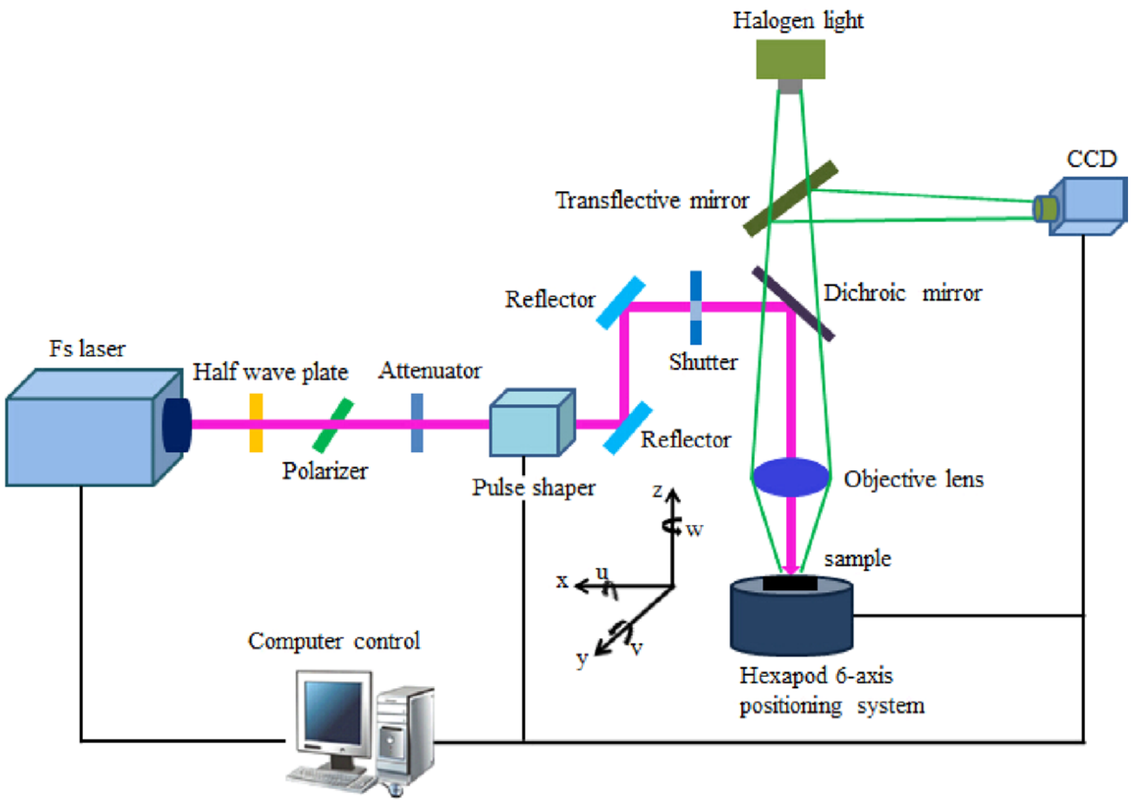

Figure 1. Schematic of experimental setup used for micromachining with shaped pulse trains.

could be continuously adjusted by combining a half-wave plate with a polarizer. The number of pulse bursts (irradiation time) was precisely controlled by using an electromechanical shutter. The fs laser pulse was temporally shaped to be a pulse train with an accurate pulse delay by a pulse shaper (BSI MIIPS BOX 640). The pulse trains used in the experiment consisted of two sub-pulses, and the pulse separations ranged from 0 fs to 5 ps. The laser beam was directed into a plano-convex lens with a focal length of 100 mm and focused at a normal incidence onto the sample which was mounted on a six-axis motion stage (M-840.5DG, PI, Inc.) with a positioning accuracy of 1 μm in the x and y direction and 0.5 μm in the z direction. The samples used in our experiments were Al (10 × 10 × 1.0 mm, purity: 99.95%, polycrystalline) and Cu (10 × 10 × 1.0 mm, purity: 99.99%, polycrystalline) with a single side that was mechanically polished.

3. Results and discussion

Firstly, the single pulse effects on fs laser ablation of Al were studied by varying the laser energy fluence and laser pulse number. Figure 2 shows the dependence of the nanoripple's period on the laser pulse number for various laser energy fluences. The periods of the ripples were measured by reading the distance between two neighboring nanostructures estimated from SEM pictures. We found that the ripple's period gradually increases as the laser energy fluence increases, while the ripple's period almost remained the same with increasing laser pulse number. We believe it is the electron density generated at the surface of the bulk material

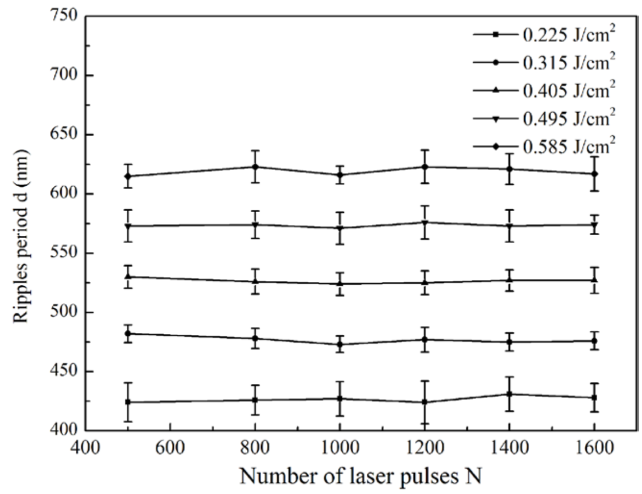

Figure 2. Nano-ripple's periods as a function of laser pulse number for various pulse energies.

that decides the ripple's period, and the electron density is proportional to the laser energy fluence. Chakravarty *et al* have confirmed that the ripple's period is relatively larger when the electron density is high [18]. High laser fluence can induce more free electrons and consequently lead to a large ripple period. As to the effects of the laser pulse number on a ripple's period, we suggest that a ripple is induced at the first several pulses (the number of laser pulses depends on the laser energy fluence; at certain high values for the laser energy fluence, only a single pulse is involved). Once the ripple structure is formed, its period is determined. For subsequent pulses, the electric field is enhanced near the

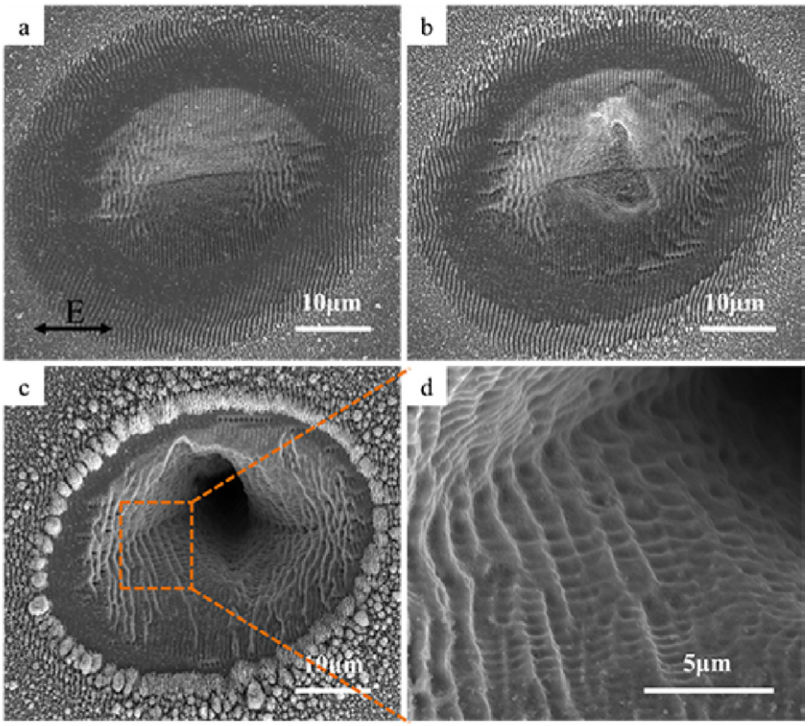

Figure 3. SEM images of ablation morphologies on Cu surfaces with a laser fluence of 2.02 J cm^{-2} and laser pulse numbers of (*a*) 500, (*b*) 1000 and (*c*) 2000. (*d*) is the magnified image of the marked part in (*c*). The double-headed arrow (E) represents the polarization vector of the laser beam.

initially formed structure, and the ripple structure becomes deeper under the influence of localized static electric fields on the surface (due to the near field effects). The subsequent pulses enhance the ripple structure but don't change the ripple's period. This agrees with Sakabe *et al* who have demonstrated that even if more pulses are irradiated on the ripple structure produced by the first several shots, the period of the ripple structure never changes [19].

Then, we performed an evolution of morphologies on the surface of Cu with a single pulse. The results under different laser pulse numbers are shown in figure 3. Figure 3(*a*) shows a ripple in the peripheral area of the irradiated spot that is just beginning to appear, while the ripple in the center area has already been slightly damaged. This is because the fs laser used in the experiment has a Gaussian spatial intensity profile. Compared with figure 3(*a*), the ripple in figure 3(*b*) in the peripheral area is more obvious, as the interaction time of the incident light with the material is prolonged and more ablated debris was deposited on the edge. This mechanism has been demonstrated by Chakravarty who reported that the molten material gets rippled [20]. The ripple's orientation in figures 3(*a*) and (*b*) is perpendicular to the laser polarization vector, which is indicated with a double-headed arrow (E) in figure 3(*a*).

When the laser pulse number arrived at 2000, an obvious crater appeared in the center area of the irradiated spot, as shown in figure 3(*c*). The ripples in the peripheral area were covered by the ablated debris. However, interlaced ripple structures formed on the wall of the crater. A magnified image of the structure is shown in figure 3(*d*). Two kinds of differently oriented ripples appeared on the wall of the crater. The period of the ripple with an orientation perpendicular to the laser polarization vector was about 1160 nm, while the ripples whose orientation was parallel to the laser polarization vector has a period in the range of 250 to 750 nm. We suggested that the vertical ripple was firstly formed under the irradiation of a linearly polarized laser beam. Then the orientation of the scattered fields which came from different scattering sources would change with the laser sequentially irradiated on the surface of Cu because of the interaction of subsequent laser beams with the formed ripple structure. The scattered fields interacted with each other to form the ablation crater. The diffraction of the ripples induced secondary ripples. Therefore, once one kind of ripple was engraved on the wall of the carter, it intervened with the subsequent light wave and made the incident laser light completely or partially overlay on the ripple. Thus, it appeared as the interlaced ripple structure on the wall of the crater. This indicates that the surface morphology strongly depends on the pulse number [21, 22].

We also studied the scan velocity and pulse separation effects on the quality and morphology of periodic ripples which induced by a fs laser pulse on a Cu surface. On this basis, we obtained different shapes and morphologies of ripple structures. Figure 4 depicts scan velocity impact on the shape and morphologies of the ripple structure with a laser energy fluence of 4.01 J cm^{-2} in a single pulse regime. All of the ripples have an orientation perpendicular to the laser polarization vector indicated with a double-headed arrow (E) in figure 4(*a*). Generally speaking, the morphology of the ripple structure becomes fuzzier with the increase of scan velocity. This phenomenon can be explained by the inter-pulse feedback mechanisms [23]. The sample would receive

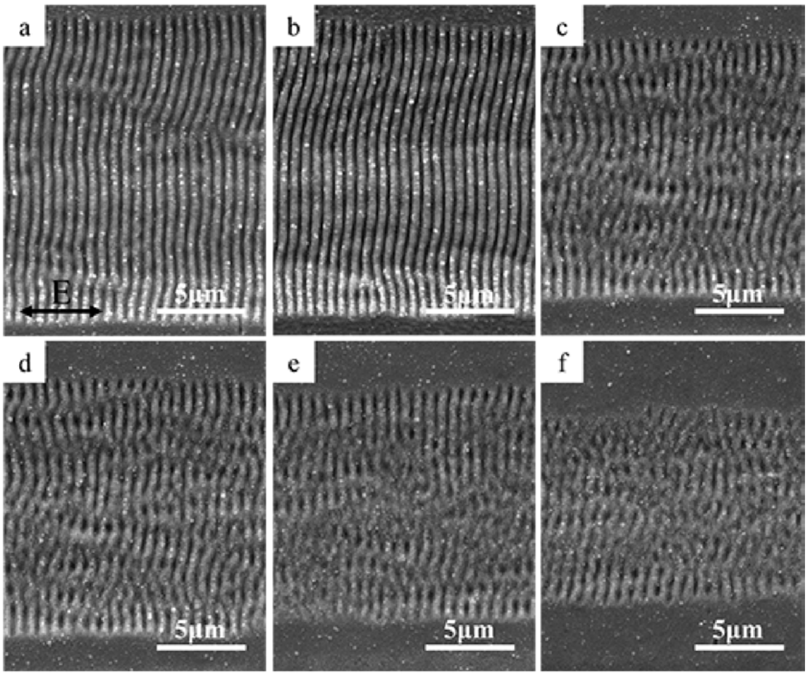

Figure 4. SEM images of morphologies on Cu surface induced by a femtosecond laser with a fluence of 4.01 J cm^{-2} and scan velocities of (*a*) 50 μm s^{-1}, (*b*) 100 μm s^{-1}, (*c*) 150 μm s^{-1}, (*d*) 200 μm s^{-1}, (*e*) 250 μm s^{-1} and (*f*) 300 μm s^{-1}. The double-headed arrow (E) represents the polarization vector of the laser beam.

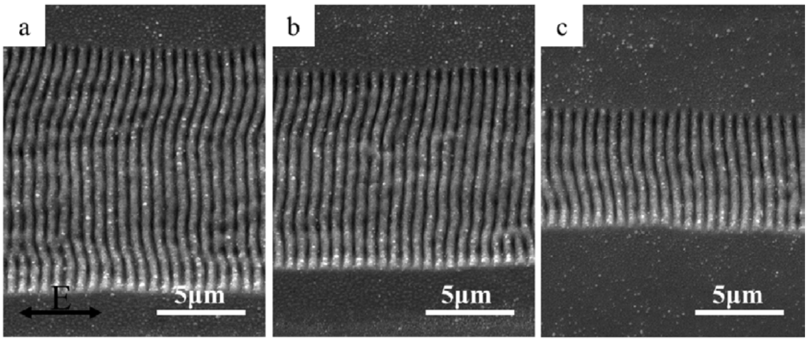

Figure 5. SEM images of morphologies on Cu surface induced by pulse trains with scan velocity of 100 μm s^{-1} and pulse separation of (*a*) 50 fs, (*b*) 100 fs, (*c*) 150 fs. The double-headed arrow (E) represents the polarization vector of the laser beam.

more pulses when the scan velocity is slower. The initial ripples which are induced by the previous pulse have a feedback mechanism that can lead to laser spot overlap, and this contributes to the formation of a ripple structure. So, the ripple has a clear morphology in the slower scan velocity regime. Conversely, the ripple has an ambiguous morphology when the scan velocity is high. Therefore, the morphology and quality of a ripple structure are closely related to the laser scan velocity [24].

Figure 5 shows the pulse train's impact on the ablation shape and morphology of the ripple structure on Cu. Pulse trains consist of double pulses in a train with a sub-pulse energy distribution of 1:1. The pulse separation between sub-pulses in a train can be precisely controlled by the pulse shaper (BSI MIIPS BOX 640). The total pulse energy fluence per trains is 4.01 J cm^{-2} and the scan velocity is 100 μm s^{-1}. We found that the ripple structures have a clear morphology when the scan velocity is 100 μm s^{-1}, as shown in figure 4(*b*), so we selected the scan velocity of 100 μm s^{-1}. In general, the ablation size becomes smaller and the quality gets better with the increase of pulse separation. This implies the capability of shaped pulse trains in controlling the ablation shapes. The free electrons' recombination has strong influence on the free electrons' density [25]. The free electrons' recombination exacerbates when the pulse separation is prolonged, which will decrease the seed electrons during the subsequent pulse in the same pulse trains. Therefore, the density of laser induced free electrons reduces with the increase of pulse separation, which will further result in a smaller ablation shape. Compared with figures 4 and 5, we found that the quality of the ripple structures obviously improved and their sizes can be well controlled by pulse trains. Thus, we conclude that the pulse train is more effective in controlling ablation shape and quality compared with a single pulse. This method may have wide applications in photonics, biochemical sensing, optoelectronics, biomedicine and other fields.

4. Conclusion

In conclusion, we have performed a series of experiments to study the morphology and related mechanisms of induced nanostructures on metal. The main findings are as follows: (1) A ripple's period increases with the increasing of the laser fluence and has a weak correlation with the pulse number. (2) Interlaced ripple structures formed on the wall of the crater with a single pulse ablation of Cu, which attributes to the incident laser light, completely or partially overlay on the ripple that was induced by the previous pulses. (3) The ripple has a clear morphology when the scan velocity is slow as a result of an inter-pulse feedback mechanism. (4) The pulse train is more effective compared with the single pulse in controlling a ripple's shape and quality by adjusting pulse separation.

Acknowledgments

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